

Particles as Topological Kinks

From the Phi-MDL Field to the Standard Model Spectrum

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UGP Physics Tutorial Series

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Prerequisites (read first): *The Φ_{MDL} Field: From Discrete Certificate to Continuum Physics · How the Standard Model Quantum Numbers Emerge · From the Arithmetic Substrate to Particle Masses*

Next tutorials: *New Physics Predictions: What GTE Predicts Will and Won't Be Found · Falsifiability: How to Test the GTE Framework*

See also: *Forces, Interactions, and the V–A Structure · The Levels of the Theory*

1 What Is a Particle in GTE?

The one-sentence answer

In the GTE framework, a particle is not a point object inserted into a field as a separate ingredient — it is a stable, localised *twist* in the Φ_{MDL} field itself, where the field transitions from one vacuum state to another without being able to unwind.

In the Standard Model, elementary particles are fundamental objects defined by their quantum numbers (charge, spin, colour). The SM writes down a Lagrangian and adds the particles as separate ingredients; their masses are free parameters